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RESEARCH ARTICLE

App Deconfliction: Orchestrating Distributed, Multi-Agent, Multi-Objective Operations for Power Systems

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ABSTRACT Advanced distribution systems need to integrate and orchestrate intelligent subsystems and grid-edge devices that are increasing both in number and sophistication while also serving multiple system-level objectives such as resilience, decarbonization, equity, and profitability. A modular platform-based approach to distribution system operations technology enables operators to deploy a tailored set of best-of-breed algorithms and applications. Combined with the parallel deployment and control of intelligent grid-edge and internet of things devices, this creates a complex distributed-control environment with applications that span ownership boundaries. Conflicts can emerge between applications that want to control overlapping sets of device setpoints. We propose a formalized approach to resolving these conflicts that can be applied when integrating new algorithms or developing customized solutions. A Deconfliction Pipeline is inserted between the device-controlling applications and the device protocol converter, which transmits control setpoints from the operations platform to the devices. The Deconfliction Pipeline executes a process that sets up, solves, and acts on a formally defined deconfliction problem. The deconfliction problem can be solved using a combination of rules and heuristics, application engagement, and optimization. We demonstrate how a few of the most basic solution strategies can be used to orchestrate harmonious behavior between a pair of simple applications with conflicting greedy optimization objectives.

INDEX TERMS Advanced distribution operations, power system management, solution design, system architecture.

I. INTRODUCTION

A. BACKGROUND

Ongoing decarbonization and grid modernization efforts are rapidly transforming electric distribution networks, with significant increases in penetration levels of distributed energy resources (DERs), including distributed generation, battery storage, electric vehicles, and controllable customer-owned resources [1]. Simultaneously, numerous new data streams are available from grid-edge devices, including advanced metering infrastructure (AMI) systems, internet of things (IoT) data, smart inverter controllers, intelligent reclosers, and micro-phasor measurement units (μPMUs). These investments in grid-edge devices and advanced applications are additionally redefining operations in distribution control rooms [2]. Historically, most devices were manually controlled by field crews (coordinating via phone with the control room dispatcher) or followed local control settings (based on direct feedback control or a pre-programmed hourly schedule). However, distribution automation projects

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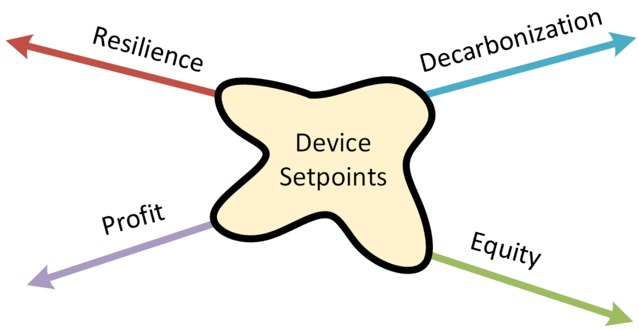


FIGURE 1. Different objectives competing to control device setpoints.

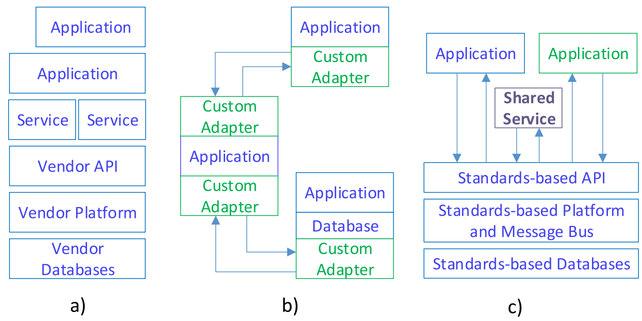


FIGURE 2. Alternatives to integration of data and power applications [8].

are shifting operations to be driven by advanced power applications taking direct control of devices without human intervention [3]. However, this transformation introduces a new set of challenges related to data integration and application interoperability previously not seen in distribution networks [4]. One such emerging issue is the conflict between advanced applications, which will be the focus of this work.

The introduction of advanced distribution management systems (ADMS) and distributed energy resource management systems (DERMS) into control rooms adds a suite of new advanced power applications with competing objectives, such as resilience, decarbonization, profitability, and equity [5], [6], [7]. Individual applications may align with one or more of these objectives but any set of functional applications cannot be expected to organically balance these competing objectives without system-level orchestration. This intrinsic tension is illustrated in Fig. 1. For example, consider a resilience application and a market application that are both controlling a battery energy storage system (ESS). The resilience of a system could be enhanced by maximizing the available energy for backup (avoiding the event) and minimizing the outage duration (recovering from the event). Similarly, the market application could use the same assets to maximize profit by minimizing system losses and avoiding energy purchases during high-priced periods. Depending on system conditions, these two applications could issue directly conflicting setpoints to charge and discharge the ESS simultaneously. Multiple apps attempting to control the same device setpoint creates a conflict that must be mitigated to avoid unintended system behavior.

B. EXISTING ARCHITECTURE AND NUMERICAL METHODS

Existing studies present some conceptual approaches towards enabling the integration of advanced applications [8] as shown in Fig. 2. One such approach includes a vertically integrated platform maintaining a suite of well-orchestrated applications using the vendor's proprietary data stack (as shown in Fig. 2a). However, the addition of new applications to such a suite would require reorchestration among the existing applications, which would not only lead to high costs but can also quickly become unmanageable [8]. An alternative to these architectures is a modular platform that facilitates the integration of custom applications either through custom adapters (Fig. 2b), which can be time-consuming and expensive to maintain, or through standards-based interfaces (Fig. 2c) which enables cost-effective development and deployment of advanced applications [9]. However, such applications may have overlapping spans of control and would interact with the operation of the other applications when integrated. Therefore there is a need for a systematic approach to resolving such conflict. Such an approach, powered by numerical methodologies, can ensure independent applications operate together in a compatible and, ideally, constructive way.

Within either a single application or a custom orchestration integration, there exist several sets of numerical methods that can be used to resolve conflicts between application objectives. Formulations of common ADMS and DERMS objective functions, constraints, and their use are reviewed exhaustively by [10], [11], [12], [13], and [14]. The first existing method for combining applications is formulation of a global optimization problem incorporating the objectives of each application, which can be solved through optimal power flow (OPF) techniques. If sufficient computational resources, communications infrastructure, a reliable network model, and a single, quantifiable, comprehensive system objective are provided, OPF can assign ideal setpoints directly [15], [16]. Efficiencies in problem formulation can make OPF solutions more computationally tractable [17], [18]. The second common approach to combining conflicting objectives is multi-objective optimization (MOO) techniques, which keep all optimization objectives separate and form a set of optimal solutions known as the Pareto-front. Pareto-front MOO is widely used in optimization of microgrids [19], [20], [21], [22], [23], [24], [25], [26] and distribution systems [27], [28], [29], [30], [31], [32]. The third alternative is multi-criteria decision making (MCDM), a branch of operations research designed to resolve conflicting qualitative and quantitative criteria under high uncertainty [33]. Numerous formulations for distribution system applications exist, including those based on the analytical hierarchy process (AHP) [34], [35], [36], [37], [38], [39], [40], simple multi-attribute rating technique exploiting ranks (SMARTER)

[41], [42], [43], and technique for order of preference by similarity (TOPSIS) [23], [26], [44], [45], [46], [47].

However, direct implementation of the numerical methods listed above face several challenges. The first is the requirement that all applications share their objective function formulation, which may be unacceptable to some ADMS vendors and developers of ''black-box'' software using novel control formulations that represent their commercial competitive advantage. Without full knowledge of all objectives, constraints, and solution variables used by all applications on the same distribution network, it is not possible to formulate a numerical problem to find globally optimal device setpoints. Furthermore, even if all application parameters are known, direct implementation of any of the methods listed above requires development of a set of custom adapters for each application, similar to the structure shown in Fig. 2 b). Introduction of a new application after completion of the previous application integration effort would require development of another custom formulation and set of software adapters.

C. CONTRIBUTIONS

This paper introduces an architecture and set of numerical methodologies that can be used to deconflict setpoint commands issued by separate power applications. The architecture is based on the concept of a Deconfliction Pipeline that uses standards-based messaging and interfaces to intercept controllable device setpoint requests, identify conflicting setpoints using a Conflict Matrix, and determine an acceptable set of compromise values forming the Resolution Vector, which is finally converted to device-specific control messages using the appropriate communications protocol. The architecture and methodologies are not only scalable across centralized, distributed, and decentralized paradigms, but also not dependent on any specific vendor platform or communications protocol. Furthermore, the methods introduced are applicable to ''black-box'' proprietary applications, ''naive'' power applications that repeatedly issue a conflicting setpoint request, and ''intelligent'' power applications that share deconfliction-related parameters in a cooperative manner.

This paper makes the following contributions to the space of multi-application platforms for distribution system operations:

1) A mathematical characterization of application conflict;
2) An architectural specification of the placement of a Deconfliction Pipeline entity within the structure of a distribution system managed by an operations platform;
3) A description of a modular Deconfliction Pipeline process designed to present common inputs to and accept common outputs from a deconflictor subprocess;
4) A formal definition of the app deconfliction problem;
5) A taxonomy of concepts, strategies, and approaches for solving the app deconfliction problem.

D. PAPER ORGANIZATION

The remainder of this paper is organized as follows: Section II characterizes the problem of application conflict. Section III defines application deconfliction, the deconfliction process, and the app deconfliction problem. Section IV describes solution methods that may be used individually or in combination to solve the deconfliction problem. Section V demonstrates application conflict and implements two simple methodologies for deconfliction. Finally, Section VI concludes the paper and describes ongoing and future work.

II. CHARACTERIZING APPLICATION CONFLICT

A conflict is observed in distribution operations platforms when applications with different objectives have mismatching preferred sets of device setpoints. If such collisions in desired setpoints are not correctly addressed, it might create operational issues such as 1) devices responding to setpoints arbitrarily, 2) oscillating system behavior as conflicting setpoints arrive asynchronously, and 3) applications completely failing to achieve their objective. The following sections detail different causes of application conflict, as well as the construction of a Conflict Matrix by assembling the preferred controllable setpoints from the applications.

A. CAUSES OF APPLICATION CONFLICT

Application conflict can arise from various factors, including system mandates, app objectives, collisions of control commands or physics-domain, and mismatch of setpoints. These factors are covered in further depth in the following subsections.

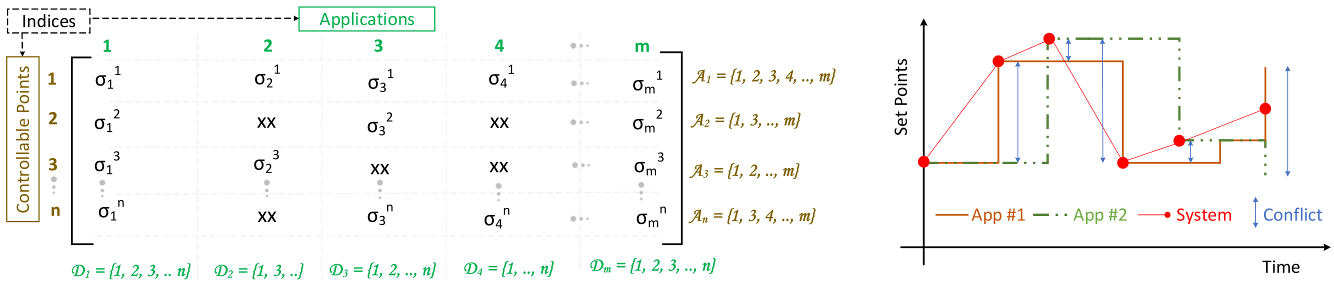
1) DISTRIBUTION SYSTEM MANDATES

Distribution system operations can be subject to a range of regulatory, customer, societal, and business mandates. Some examples include satisfying customers, maintaining reliability and profitability, decarbonizing generation, maximizing resilience, and achieving equity. These system mandates can cause application conflict when the objectives of different applications are aligned with different sets of mandates.

The composition and prioritization of this list depends on locational factors and can change over time. While system operators need to satisfy each applicable mandate, individual applications within an advanced operations platform, which are designed to perform one or more specific technical functions (such as fault location isolation and service restoration (FLISR) or volt-var optimization (VVO)), may be programmed in a way that is incidentally consistent with or intentionally prioritizing a subset of the mandates that apply to the system.

2) APPLICATION OBJECTIVES

The objective of an application is the desired outcome of application execution. The objective of an application may or may not be expressed as a formal optimization objective



(a) (b)

FIGURE 3. Representing conflict: a) A matrix representation of apps conflicting with each other while controlling a set of devices. Note that an "xx" element at a location (i, j) in the matrix represents that a i^{th} controllable point is not considered by j^{th} app; b) Example illustrating a time-varying nature of app conflict when App #1 and App #2 have different intended set points (the magnitude of this conflict is shown in blue) indeterminate behavior is observed when App #1 and App #2 send conflicting setpoints with the same time stamp as in the last setpoint update shown.

function. Both optimization-based and non-optimization-based applications have objectives that can lead to a conflict.

3) CONTROL-COMMAND COLLISIONS

When different applications attempt to control the same device by sending different control setpoints to the device, this causes a control-command collision. This is one type of application conflict. A hypothetical example of a control-command collision would occur if one application sends a "close" command to a capacitor bank to minimize line losses but a second application sends an "open" command to the capacitor to implement conservation voltage reduction (CVR) while the capacitor behavior intended by the first app remains unchanged.

4) PHYSICS-DOMAIN COLLISIONS

A physics-domain collision, a second type of application conflict, occurs when different applications issue control commands that counteract each other in the physics domain. An example of such a collision involves one application that sends commands to dispatch reactive power of batteries (to increase voltage) and another application that subsequently opens capacitor banks (to decrease voltage). These applications do not conflict directly from a device setpoint perspective, but are issuing commands that counteract each other and need to be deconflicted.

5) SETPOINTS MISMATCH

A setpoints mismatch occurs whenever the ideal value of a particular controllable device setpoint differs depending on the application perspective. Setpoint mismatches are concerned with the ideal intention of a set of applications and therefore describe both control-command collisions and physics-domain collisions.

B. CONFLICT MATRIX

A Conflict Matrix can be constructed to describe the relationship between applications and device setpoints. In a Conflict Matrix, the set of controllable setpoints in the system is arranged in a series of columns, each corresponding to

TABLE 1. Glossary for comprehending a conflict matrix.

$\mathcal{A}$	Set of indices of apps operating on a system, $m = \lvert\mathcal{A}\rvert$
$\mathcal{D}$	Set of indices of controllable points on a system, $n = \lvert\mathcal{D}\rvert$
$\mathcal{A}_i$	Set of indices of apps colliding in i^{th} controllable point, $\mathcal{A}_i \subseteq \mathcal{A}$
$\mathcal{D}_j$	Set of indices of controllable points where j^{th} app is requesting an update, $\mathcal{D}_j \subseteq \mathcal{D}$
σ_i^j	Control requested by i^{th} app in j^{th} controllable point
xx	Null or "don't-care" set point

an application (See Fig. 3a). Each column vector in the matrix describes the app's intent ($\mathcal{D}_j$). Similarly, each row vector in the matrix represents the set of competing apps for a given controllable point ($\mathcal{A}_i$). Such matrix representation establishes a decision space with dimensionality equal to the number of controllable points in the system for a given point in time. Please refer to Table 1 for the glossary associated with the Conflict Matrix shown in Fig. 3a.

An example of conflict in a single controllable point is shown in Fig. 3b. Suppose two applications control a device independently and send setpoint values simultaneously. App #1 sends five control signals during the time under consideration, whereas App #2 sends four control signals. However, the system observes six requests within the given time. Additionally, both apps can't attain their desired behavior from the system. As illustrated in Fig. 3b, applications may have different operational cycles, and conflict in a system varies with time. Thus, the Conflict Matrix is updated whenever a new set of preferred controllable set points are received from an application. The updated Conflict Matrix can be analyzed to identify and resolve possible conflicts during the deconfliction process.

C. CONFLICT MATRIX SCALABILITY

The characterization of application conflict and solution techniques described in this paper are applicable to centralized, distributed, and decentralized architectures. Centralized architectures are characterized by aggregation of communications and control networks in a single utility control room,

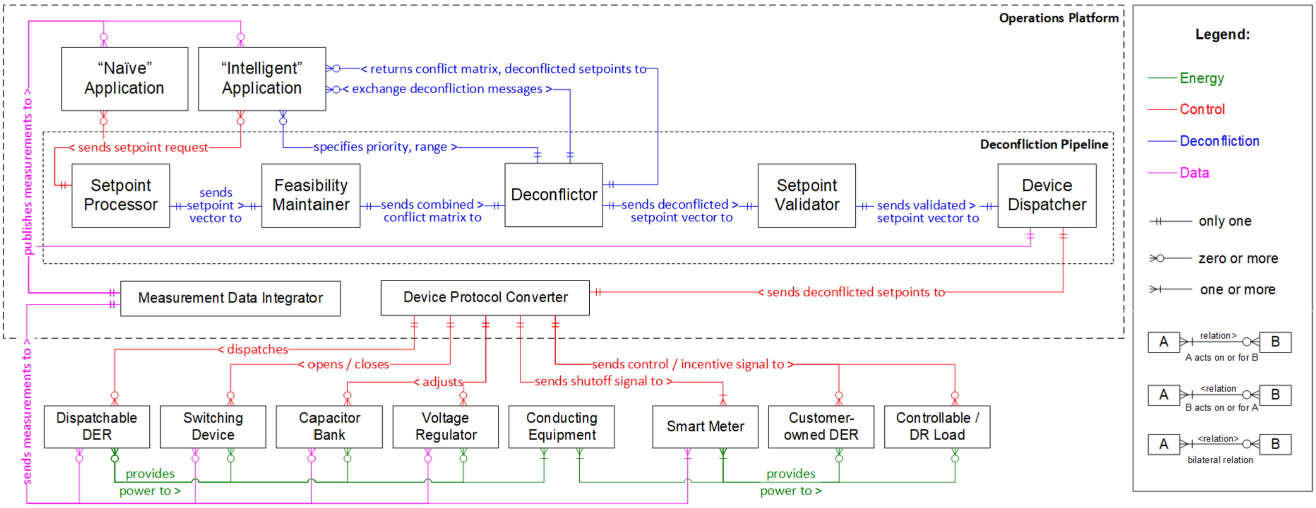


FIGURE 4. Entity-relationship structure diagram for distribution system actors illustrating insertion of the Deconfliction Pipeline between applications and controllable field devices.

with all advanced apps hosted on a centralized collection of network servers. In this context, the Conflict Matrix is built for all equipment within the utility service territory with a single global matrix containing a row for each control setpoint of each device. The Conflict Matrix can be partitioned to consider only the setpoints of controllable devices across a single feeder or the entire distribution system depending on the particular implementation.

Distributed architectures seek to divide the distribution network into independent areas with hierarchical or peer-to-peer coordination between individual distributed application agents. Distributed control areas can be defined using either the communications network structure or the topology-based approach introduced in [48]. Implementations in a distributed architecture only include in the Conflict Matrix the set of controllable device setpoints within the boundaries of that area. The deconfliction process is distributed across each distributed control area such that each distributed Deconflictor tracks the setpoints in the Conflict Matrix for that distributed area and then coordinates with other deconfliction agents to share decisions and impacts to system attributes.

In a decentralized architecture, no coordination is used between local control agents. Decentralized implementations of the Conflict Matrix compile setpoints for a single controllable device (or device group) and could potentially be performed by a microgrid controller or by the grid-edge devices themselves.

III. DECONFLICTION

This section details a systematic approach to orchestrating system operations to resolve conflicts between controllable-device setpoints produced by various applications. The approach introduces a modular software construct called the Deconfiliction Pipeline that can be implemented in centralized, distributed, or decentralized architectures using the Conflict Matrix for the entire distribution system or an individual local control area.

A. STRUCTURE FOR CONTROL AND DECONFLICTION

Within the structure of an operations platform, deconfliction of setpoints from competing applications can be achieved by inserting a Deconfliction Pipeline between applications and the device protocol converter. The Deconfliction Pipeline receives device setpoint messages, deconflicts them using one of the solution methods described in Section IV, and then publishes deconflicted setpoints that are converted to the appropriate network communications protocol (such as DNP3 and IEEE2030.5), as shown in Fig. 4, which presents a detailed entity-relationship structure diagram for classes of entities involved in application deconfliction.

A key prerequisite to developing an effective deconfliction framework is understanding the interactions between stakeholders, applications, devices, and other actors involved in power system operations. This discussion will draw on the domain of Grid Architecture [49], which recognizes that that distribution grids are not simply an electric circuit, but rather a network of structures involving control, coordination, communications, sensing, and data management. It is possible to capture these complex structures with an entity-relationship diagram, which describes classes of entities and different modes of interaction between them. Detailed industry structure diagrams for the bulk electric system, distribution system operators, and distribution networks with high DER penetration are presented by [49] and [50].

The entity-structure diagrams for distribution systems are extended down to the device level in Fig. 4, which depicts a subset of the relationships and classes of devices controllable by advanced applications. Relationships and interactions not directly related to application deconfliction (such as ownership and market transactions) are omitted for visual clarity. The diagram also omits devices that are not controllable by applications (such as manually controlled sectionalizers and fuses) and operations actions that cannot be executed through a setpoint request (such as re-phasing or cuts/jumpers made by field crews).

Figure 4 highlights four categories of relationships between the classes of controllable devices (shown at bottom) and software entities contained in a data integration platform. The first set of interactions focus on flow of energy among controllable devices, with dispatchable DERs providing real and reactive power to other devices through a set of non-controllable conducting equipment to customers supplied by individual meters. The meters and controllable assets send measurement data back asynchronously to a measurement data integrator inside the software platform, which aggregates and correlates the data for use by applications and power system operators. Applications subscribe to measurement data streams and generate device setpoints using the app's internal objective and configuration settings applied by the power system operator.

The device setpoints from each application are received by the Deconfliction Pipeline, which is expanded in further layers of detail in Fig. 4. The Deconfliction Pipeline can exchange deconfliction messages with intelligent applications as part of coordination-based or optimization-based deconfliction schemes. The Deconfliction Pipeline can also exchange deconfliction information with the power system operator, such as messages prompting the operator to take manual control action or prioritize apps if an acceptable deconfliction could not be found. The power system operator can also configure the Deconfliction Pipeline to grant an application exclusive control over certain devices or give setpoints for an application higher priority than competing setpoints from other applications. Deconflicted setpoints generated using any of the three approaches described in Section IV are provided to the device protocol conversion service(s) that converts the setpoint to the correct protocol and sends it on the appropriate communications network to the field device.

The structure diagram applies to centralized, distributed, and decentralized control architectures, with the same agents and interactions occurring inside each local control area. In distributed architectures, distributed Deconfliction Pipeline instances may coordinate and share decisions with each other using a hierarchical or peer-to-peer structure.

B. THE DECONFLICTION PIPELINE

To provide a consistent, flexible method for deconflicting applications, the concept of a Deconfliction Pipeline is introduced. The Deconfliction Pipeline is created in a standards-based platform environment and serves as the bridge between setpoints issued by applications hosted on the information technology (IT) data bus and physical devices connected to the operational technology (OT) data bus through a communication network. The role of the Deconfliction Pipeline is to intercept all setpoint requests issued by applications, identify possible conflicts, and create new deconflicted setpoints that are sent to the physical devices. This process is decomposed into the five functional blocks illustrated in Fig. 4.

1) SETPOINT PROCESSOR

The Setpoint Processor compiles setpoints from individual applications into rows in the Conflict Matrix that contain the desired value (or ''don't-care'' status) of each setpoint of controllable devices in the feeder or local distributed control area. ''Intelligent'' applications can publish the set of setpoints for all devices the app is using to achieve its individual objective(s), which provides sufficient information to construct a setpoints-row directly. ''Naive'' applications may 1) send asynchronous subsets of setpoints for controllable devices, 2) omit control commands for devices that presently have a desired setpoint, and/or 3) omit expression of ''don't-care'' status for setpoints. It is the responsibility of the Setpoint Processor to interpret the interest and intent of naive apps and complete the row, filling in any missing values.

2) FEASIBILITY MAINTAINER

The Feasibility Maintainer ensures that the Conflict Matrix includes only feasible setpoints. Setpoints are deemed infeasible if they exceed physical or imposed device limits. When infeasible setpoints are encountered, the Feasibility Maintainer may reject or modify values.

3) DECONFLICTOR

The Deconflictor is responsible for converting the set of app setpoints represented by a Conflict Matrix into a Resolution Vector as described in III-C. A wide array of methodologies may be employed by the Deconflictor as described in Section IV. The Deconflictor may exchange contextual information using the implemented platform application programming interface (API) with applications such as Deconfliction Signals (e.g., coordination or incentive signals), the current Conflict Matrix, the current Resolution Vector, App Status, and/or App Metrics.

4) SETPOINT VALIDATOR

The Setpoint Validator ensures that the setpoints described by the Resolution Vector are consistent with operational requirements (e.g., safety or regulatory). If the Resolution Vector or setpoints therein are deemed to be invalid, automatic or operator intervention may be required.

5) DEVICE DISPATCHER

The Device Dispatcher, the last stage of the Deconfliction Pipeline, is responsible for comparing the Resolution Vector to the operational values of setpoints and issuing platform-level control commands in the format of CIM difference messages or other standards-based message published onto the platform IT data bus. These messages are then converted by the Device Protocol Converter to the correct communications protocol and distributed to the controllable devices via the platform OT data bus and physical communications layer.

C. DECONFLICTION PROCESS DETAIL

The Deconflictor (as discussed in Sec III-B) serves as the key functional element towards mitigating conflict among the application preferred setpoints. Although diverse approaches can be applied to deconfliction, the overall goal of the Deconflictor is to determine deconflicted controllable-setpoints for devices. The internal functional processes involved in achieving that goal can be broadly classified according to the functional blocks shown in Fig. 5.

1) CONFLICT IDENTIFICATION

The Conflict Matrix provides a systematic way of identifying conflict among applications. The first substep of the Deconflictor is parsing the Conflict Matrix to identify setpoint mismatches. If a conflict is detected, the numerical Deconfliction Solution is triggered to execute for the current iteration.

Various applications may publish setpoints at different time intervals, the Conflict Matrix is updated by the Setpoint Processor and Feasibility Maintainer whenever a new set of setpoints is received from an application. This triggers the Deconflictor to review the Conflict Matrix for any new conflicts.

2) DECONFLICTION SOLUTION

If a conflict has been identified, the Deconflictor develops a new vector of setpoints using one of the methodologies described in Section IV, including rules & heuristics, context & cooperation, and optimization. Implementation may vary based on situational circumstances. The Deconfliction Solution process receives the Conflict Matrix along with additional information published by the applications to aid the deconfliction process, such as operational status, objective function formulations, and desired setpoint ranges. In order to facilitate some degree of awareness and cooperation among the applications, the solution process is also permitted to communicate suitable deconfliction signals (like coordination signals, incentive signals, or fitness functions) along with conflict-related information (e.g., an updated Conflict Matrix) back to the applications for coordinated decision-making, as illustrated in Fig. 5.

3) RESOLUTION

The resolution process involves assembling the deconflicted solutions into a Resolution Vector as shown in Fig. 5. The Resolution Vector is then communicated to the dispatcher through the pipeline and used to issue device-level control commands for dispatch. This Resolution Vector can also be communicated to applications (as shown in Fig. 5) to inform them of the final deconflicted setpoints sent to devices. This information can be leveraged by applications that use learning-based algorithms to make more informed decisions or schedule-based applications (e.g., day-ahead DER scheduling) to update their inputs for the next time step.

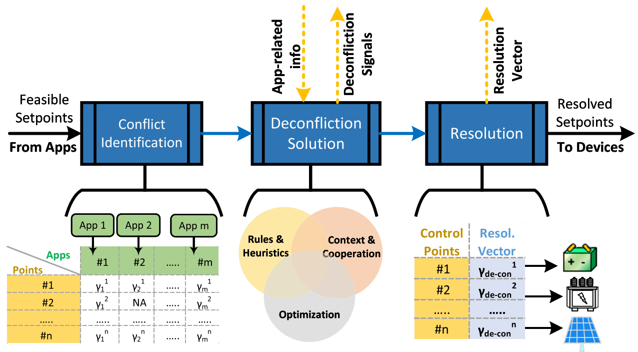


FIGURE 5. Conceptual overview of the deconfliction process.

IV. DECONFLICTION METHODOLOGIES

A variety of deconfliction methodologies can be used to solve the deconfliction problem, ranging in complexity from deference to a single app to modal multi-objective optimal power flow integrating objectives and constraints from the system-level and multiple apps. The following concepts that describe relationships between devices and apps can be used individually or in combination to develop deconfliction strategies.

- **Exclusivity**: Each setpoint is determined directly by a specific app.
- **Priority**: Each setpoint is determined by the highest-priority app actively requesting a valid setpoint.
- **Preference**: Each setpoint is a function of setpoints requested by apps and a weight assigned to each app's preference.
- **Cooperation**: Applications share additional data in a standardized manner to facilitate a co-optimal solution.
- **Coordination**: A centralized mechanism is used to define setpoints.
- **Compromise**: Setpoints are selected by a deconfliction methodology.
- **Command**: Setpoints are determined by a human operator, relay, or other entity with the authority to override applications.

A. LEVELS OF DECONFLICTION

The aforementioned concepts correspond to different levels of modeling detail, namely the device-level, app-level, system-level, or a combination thereof, as shown in Fig. 6. Each level in the centralization hierarchy provides insight into a particular approach to solving the deconfliction problem. The lowest level on which deconfliction can be performed is at the device-level, with conflicting setpoint requests evaluated for each controllable device. In this context, the focus will be on distribution assets that can be adjusted through a supervisory control and data acquisition (SCADA) system or other communication networks. The app-level approach to deconfliction aims to enable the applications to cooperate in a coordinated fashion to resolve a conflict. The app-level coordination approach can reduce the complexity of the deconfliction problem by utilizing applications' intrinsic

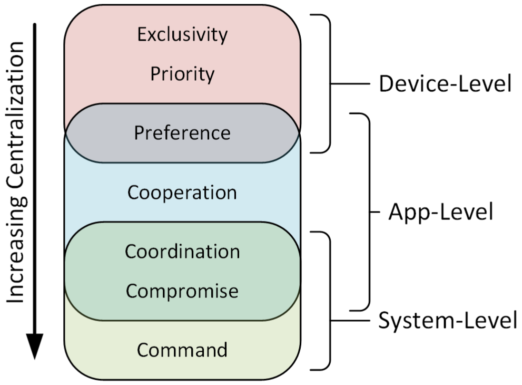


FIGURE 6. Deconfliction concepts according to level of centralization.

capabilities and can provide informed decisions through contextual information exchange, thereby making it scalable and interoperable for a diverse set of applications. In the context of system-level deconfliction, optimization provides the ideal framework to achieve overarching outcomes. System operators know their objectives and have access to system-level measurements and the ability to take system-level control actions.

B. METHODOLOGIES

Three different deconfliction methodologies are described in detail. Note that while each method is driven by consideration of a single level, each method, at its best, draws from information at each level.

1) RULES AND HEURISTICS

The first method to deconfliction is through a rule-based framework focused on asset health and heuristics to determine an acceptable balance between competing setpoints identified in the Conflict Matrix. Individual devices and system properties are used as the basis for identifying conflicts between applications. Rules-based deconfliction can be performed in a centralized, distributed, or decentralized manner with an overall goal of protecting the health of assets and ensuring safe operation of the power system, with enforcement of additional technical, economic, environmental, and social decision criteria.

The deconfliction framework combines MCDM heuristics with a set of strict rules based on the physical characteristics and limitations of each asset. The device-based rules are focused on not only ensuring that applications do not violate the setpoint limits, but also that the cumulative impact of deconfliction results do not impact asset lifespan (such as limiting the number of transformer tap steps per day or discharging a utility-scale lithium battery to maintain state of charge below 90 percent during hot weather). After applying the initial set of device-based rules, the deconfliction framework considers the priority of each application and preference given to each setpoint. The preference and priority can be used as weighting criteria for a heuristic solution for the setpoint Resolution Vector. Relationships between apps and devices, such as exclusivity, priority, and preference are handled as pass-through properties. Power system operators could conceptually specify individual devices that an application is granted control of, as well as specific devices for which an application is granted prioritized setpoint requests. Likewise, application-specified preferences for setpoints are passed down with the given weighting applied for the setpoints of each individual device. Depending on the requirements of the deconfliction implementation, the numerical component of the heuristics solution can range in sophistication from selection of the weighted centroid of conflict penalty metrics to a full utility-theory approach considering key device-level and system-level decision criteria using a discrete MCDM method such as SMARTER.

2) CONTEXT AND COOPERATION

The context- and cooperation-based approach aims to enable applications to mitigate conflicting operational preferences through mutual coordination. This can be achieved through exchanging contextual information (deconfliction signals) among the applications or through additional system-driven incentives. Context and cooperation among applications can be realized through several methods.

Game theory presents one such approach towards cooperative decision-making through strategic interaction models. Game theory-based mechanisms can be leveraged where individual applications with greedy objectives can be considered players in a cooperative game and reducing the degree of conflict among the players can be considered as the benefit of a coalition. Some existing works have used such cooperative game methods for solving multi-microgrid [51] and virtual power plant management problems [52]. In a similar context, suitable deconfliction-related incentives could be designed to motivate applications to update their operational strategies to reduce conflict.

An alternative to game theory-based approaches are MCDM techniques that focus on combining conflicting goals and evaluating multiple conflicting criteria to achieve informed decision making. In general MCDM methods such as preference ranking organization method for enrichment evaluation (PROMETHEE), elimination end choice translating reality (ELECTRE), and TOPSIS are well suited to rank the operational preference of the applications given the context of their decision. However, such approaches require some mechanism to allow the applications to use the resulting rankings to reassess their decision and cooperate. One example approach for MCDM would be TOPSIS for group decision-making. Each app presents its own objective and attributes to the decision matrix for TOPSIS. These decision matrices are then aggregated and ranked in similarity to each other which identifies which candidates are near consensus and those that need to be deconflicted by each of the cooperating apps. Fu et al. introduce the concept of group satisfaction using TOPSIS for group decision-making and highlight the importance of a feedback mechanism to help decision-makers cooperate and improve satisfaction and consensus [53].

3) OPTIMIZATION

Optimization-based approaches can be flexible in order to accommodate different types of information that can be produced by distributed applications. At a minimum, applications provide only setpoints for devices that they aim to control. At the other end of the complexity spectrum, applications can provide their full internal objectives and constraints to the Deconflictor. System operators may also assign priorities or preferences to different applications, or even set system-level or aggregate objectives and constraints.

As an example, if each application provides only its desired setpoint, optimization can resolve conflicts between app setpoints, e.g., by minimizing the distance between the Resolution Vector and the set of app setpoint vectors. In this case, operator priorities would manifest as weighting factors for the terms of the objective function.

On the other hand, applications may provide their internal objectives, in which case we can use multi-objective optimization techniques to form a single system-level objective. This objective can be augmented to a system-level optimization problem through the aggregation of constraints provided by different applications with system-level constraints (e.g., security constraints). By solving such a system-level optimization problem we can obtain unique setpoints for each device and eliminate conflicts.

V. DEMONSTRATION

In this section, we demonstrate deconfliction by applying techniques identified in Section IV. The demonstration is implemented within GridAPPS-D, an open-source platform providing a service-oriented architecture (SOA) for integrating advanced distribution management system (ADMS) applications for operations and planning. First, we implement two competing applications with diverging objectives and discuss the baseline conflict. Then we present simulation results on how different deconfliction methods can help to mitigate conflict within the system.

A. TEST SYSTEM

The 9500-node test feeder[1] is selected for the demonstration. The common information model (CIM) representation of the test feeder model is maintained within a GridAPPS-D Blazegraph database and is publicly available in extensible markup language (XML) format [55].

B. SCENARIOS

For the implementation, custom load and PV profiles are used in the simulation (See Fig. 7). For demonstration purposes, a 15-min period ($\tau = 0.25$ hour) is regarded as the control cycle for applications, and a day-long simulation is executed (i.e., 96 discrete time steps). The two outage conditions are simulated to demonstrate the time-varying conflicts and different deconfliction methods. Outage #1 occurs during the evening (i.e., 6:00 PM to 9:00 PM) and Outage #2 occurs during the afternoon (i.e., 2:00 PM to 5:00 PM). The competing apps will dispatch batteries in response to the given scenario (load and PV profile) and simulated grid outage conditions.

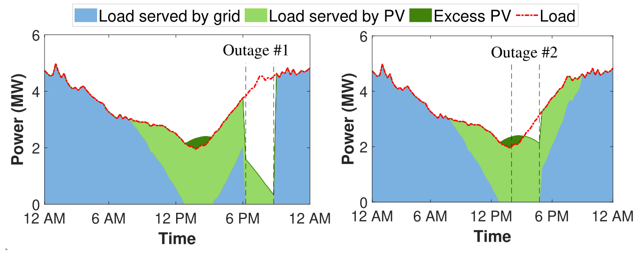


FIGURE 7. Scenarios for demonstrating deconfliction methods. Two grid outages are simulated; Outage #1 represents evening (6:00 PM to 9:00 PM) and Outage #2 represents afternoon (2:00 PM to 5:00 PM) outage. Note that these outages are considered to be on different days and simulated independently.

C. COMPETING APPS

We define a pair of greedy apps with diverging objectives to illustrate conflict. For demonstration purposes, resilience and decarbonization apps are implemented. For simplicity, only the energy balance equations are considered to achieve the app objectives (i.e., no consideration of network loss and operating constraints).

1) RESILIENCE APP

The system's resilience can be enhanced by 1) maximizing the available energy (proactive action) during the normal state [56] and 2) maximizing the load served during the emergency state [57]. Fig. 8 shows the workflow for the resilience app. The app aims to charge the batteries to their maximum level and increase available energy within the system. The required power for charging a battery (P^c_{batt}) is calculated using the current state of charge (SoC), battery capacity, and forecasted values of renewable resources (ΣP_{ren}) and loads (ΣP_{load}). If renewable generations are insufficient, the app will charge the batteries using power from the grid (P_{sub}) as shown in Fig. 8.

The resilience app enters an emergency state when there is a scarcity condition (i.e., $\Sigma P_{ren} + P_{sub} < \Sigma P_{load}$). During emergencies, the app will discharge the batteries (P^d_{batt}) and decide if emergency actions are required. The emergency actions include dispatching fossil-fueled distributed resources such as diesel generators or load curtailment to maintain the power balance in the network.

We define the following two metrics to quantify the resilience of a grid.

[1] The original IEEE 8500-node is modified to be supplied from multiple substations. Additional switches are added for possible load transfer within feeders, and several DERs (rooftop photovoltaics (PVs), storage, and different combustion generators) are added [54].

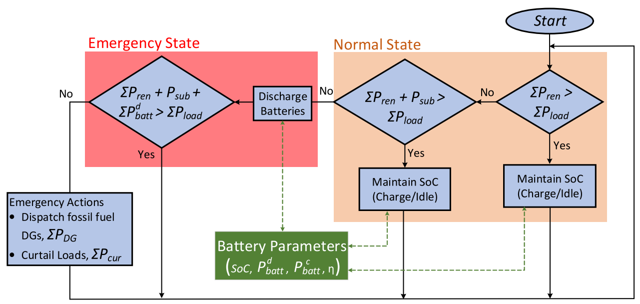


FIGURE 8. Workflow for the resilience app.

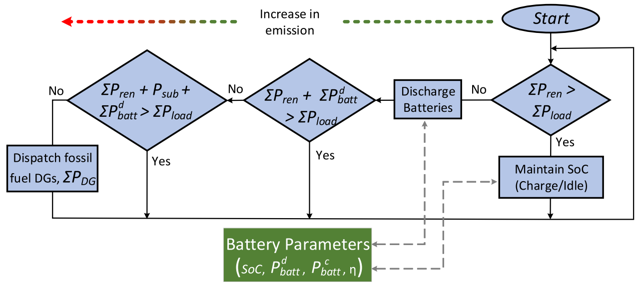


FIGURE 9. Workflow for the decarbonization app. The emissions increase as the app acquires dirty generations (grid power and fossil-fueled DGs).

- Average SoC ($\mathcal{R}_1$): This metric calculates the average SoC of the batteries. The higher value of average SoC signifies the resilience app's ability to keep an adequate reserve for the emergency grid condition.
- Storage utilization ($\mathcal{R}_2$): This metric calculates the total energy from batteries utilized during the outage period, measured in kW-hr.

2) DECARBONIZATION APP

The decarbonization app seeks to cut emissions by using sustainable energy to meet the demand to the greatest extent possible [58]. One way to implement a decarbonization app is to solve the economic dispatch problem by taking environmental concerns into account; the emission-constrained economic dispatch problem considers emission control as one of its operational objectives [59], [60]. Such bi-objective optimization problems could be solved using various techniques such as weighted sum and Pareto front; however, the focus of this paper is not to develop a robust economic dispatch model. Thus, we design a basic decarbonization app that aims to 1) minimize emissions by prioritizing the available renewable resources while supplying feeder demand, and 2) maximize renewable utilization by storing excess power in batteries. In this app, the grid power and fossil-fueled distributed generators (DGs) are considered dirty generation sources.

Fig. 9 shows the workflow for the decarbonization app. First, the app utilizes renewable energy resources to supply the load and uses surplus power to charge the batteries. If surplus renewable energy is unavailable (i.e., $\Sigma P_{ren} < P_{load}$), batteries are discharged to supply the demand. The grid power is only used if there is an insufficient reserve in the batteries (ΣP^d_{batt}). The lowest priority is given to fossil-fueled DGs such as diesel gen-sets, micro-turbines, and LNG engine generators within the feeder and are procured only for meeting extreme needs (i.e., when $\Sigma P_{ren} + P_{sub} + \Sigma P^d_{batt} < \Sigma P_{load}$).

We define the following two metrics to measure the performance of the decarbonization app.

- MW-hr reduction in dirty generation ($\mathcal{D}_1$): This metric evaluates the total reduction in energy procured from the grid and fossil-fueled DGs. It measures the capability of the decarbonization app to defer the use of dirty generation.
- MW-hr of renewable utilized ($\mathcal{D}_2$): This metric calculates the total renewable energy utilized during a day-long simulation. It measures how well the decarbonization app can utilize renewable energy resources to supply the load and use surplus power to charge the batteries.

3) BASELINE CONFLICT

When the resilience and decarbonization apps work independently to achieve their own objectives, they might generate conflicting solutions. For example, during normal operating conditions, the resilience app will dispatch a signal to charge the batteries ($P^c_{batt} \leq \overline{P_{batt}}$) in order to increase the available reserve in the system. In contrast, to cut emissions by eliminating grid power, the decarbonization app will communicate the discharge setpoint ($P^d_{batt} \leq \underline{P_{batt}}$) to batteries. In such a case, a maximum conflict is observed if batteries are charged/discharged with rated power ($\overline{P_{batt}}$ or $\underline{P_{batt}}$).

An intermediate conflict could also be observed during times when apps have overlapping objectives. For example, during peak PV generation, the decarbonization app charges batteries with limited power ($P^c_{batt} \leq \overline{P_{batt}}$) using the surplus power from renewable resources only. Conversely, the resilience app charges batteries with the rated power ($P^c_{batt} = \overline{P_{batt}}$) using both renewable resources and grid power.

D. DECONFLICTION IMPLEMENTATION

The concept of exclusivity and compromise is used to resolve the conflict between the two competing apps previously discussed. Note that these concepts are intended for demonstration purposes only and should not be considered a comprehensive solution to the deconfliction problem.

1) EXCLUSIVITY

The exclusivity concept is implemented where the resolved setpoint is determined directly by an app. Separate implementations of this strategy are used for the two apps. In resilience exclusivity, the setpoint requested by the resilience app is considered the resolved setpoint. Similarly, in decarbonization exclusivity, the resolved setpoint is the solution of the decarbonization app.

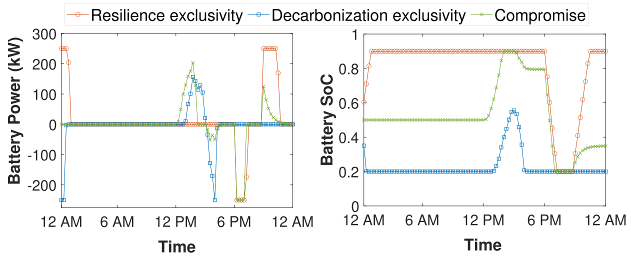


FIGURE 10. Deconfliction solutions for outage scenario #1. Resolved setpoints after deconfliction (left) and SoC after setpoints dispatch (right).

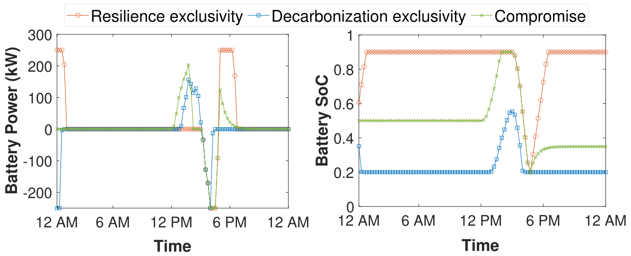


FIGURE 11. Deconfliction solutions for outage scenario #2. Resolved setpoints after deconfliction (left) and SoC after setpoints dispatch (right).

2) COMPROMISE

The compromise concept for deconfliction evaluates the average value of the setpoints generated by the decarbonization app and the resilience app as a solution.

E. RESULTS

The performance of deconfliction methods under two grid outage conditions is presented herein. The setpoints (P^c_{batt} or P^d_{batt}) and the resulting SoC are calculated using each deconfliction strategy. As obtained from the deconfliction solution, the resolved setpoints are dispatched to the devices, and the apps use the updated system states to find the setpoints for the next step.

1) DECONFLICTION SOLUTIONS

Figs. 10 and 11 show the deconfliction solutions for outage #1 and outage #2, respectively. When the resilience exclusivity method is implemented, batteries are charged to their full capacity and are only discharged during an emergency. For example, in response to outage #1, batteries start to discharge around 6:00 PM to supply the feeder demand. Note that if battery power is insufficient, emergency actions (DG dispatch and load curtailment) are carried out. Once the outage period is over, the batteries start to charge to maintain the system reserve (see Fig. 10). On the other hand, the decarbonization exclusivity method allows batteries to charge only if there is surplus power from renewables (i.e., during peak PV generation, $\Sigma P_{ren} > \Sigma P_{load}$). For example, the batteries start to charge around 1:00 PM using surplus power from the renewables. Once the excess PV period is over, the batteries stop charging and discharge to reduce emissions. Because of this, the decarbonization exclusivity technique is not always able to assist the grid during outages. For example, in outage #2, the decarbonization app will only charge the battery from 1:00 PM to 3:00 PM using surplus power from the PV. It starts to discharge after 3:00 PM due to the grid outage. However, due to insufficient energy in the battery, it is exhausted sooner as compared to the resilience exclusivity case (see Fig. 11).

The compromise technique calculates the resolved setpoint by taking an average of setpoints requested by the resilience app and the decarbonization app. The compromise solution usually results in inaction due to disagreement between the two applications under consideration. Exceptions occur during emergency grid conditions and excess PV generation. This is because the setpoint requested by the resilience app and the decarbonization app cancel each other while taking an average (See Fig. 10 and 11). During the excess PV generation period, both the decarbonization app and the resilience app request setpoints to charge the batteries. However, the setpoint requested by the decarbonization app depends on the available battery capacity and surplus power, whereas the setpoint requested by the resilience app depends only on the battery capacity. So, an average setpoint during this condition will start to charge the battery with a value higher than the decarbonization exclusivity. This can be observed during the excess PV generation period in Fig. 10 and 11. With the compromise effect, the resilience app will control the decarbonization app in discharging batteries at rated power except during an outage (see compromise solution in Fig. 10 after 2:00 PM). On the other hand, the decarbonization app will restrict the resilience app from charging the batteries with rated power unless there is surplus renewable energy (See compromise solution in Fig. 11 after 5:00 PM).

2) APP PERFORMANCE

Based on the results of different deconfliction solutions discussed earlier, the metrics defined in Section V-C are calculated for both outage scenarios. Table 2 and Table 3 summarize the trade-offs for the resilience and the decarbonization apps when different deconfliction solutions are implemented.

TABLE 2. Performance of the resilience and decarbonization apps for outage scenario #1.

Deconfliction Implementation	Resilience		Decarbonization	
	$\mathcal{R}_1$	$\mathcal{R}_2$	$\mathcal{D}_1$	$\mathcal{D}_2$
Resilience exclusivity	0.807	293.475	18.738	19.039
Decarbonization exclusivity	0.227	0	19.588	19.462
Compromise	0.522	249.533	19.395	19.270

TABLE 3. Performance of the resilience and decarbonization apps for outage scenario #2.

Deconfliction Implementation	Resilience		Decarbonization	
	$\mathcal{R}_1$	$\mathcal{R}_2$	$\mathcal{D}_1$	$\mathcal{D}_2$
Resilience exclusivity	0.847	293.475	18.738	19.039
Decarbonization exclusivity	0.227	148.857	19.588	19.462
Compromise	0.486	293.475	19.395	19.270

In both scenarios, the resilience metrics ($\mathcal{R}_1$ and $\mathcal{R}_2$) have the highest value for the resilience exclusivity method and the least value for the decarbonization exclusivity method. During outage scenario #1, as shown in Table 2, $\mathcal{R}_2$ is zero when using the decarbonization exclusivity method. This is because the battery was fully exhausted before the outage scenario, as shown in Fig. 10. Similarly, $\mathcal{R}_2$ is slightly lower for the compromise solution when compared to the resilience exclusivity method. This is because, right before the outage, the battery SOC was higher for the resiliency exclusivity method. On the other hand, during outage scenario #2, $\mathcal{R}_2$ for the decarbonization exclusivity method is about half as compared to the resilience exclusivity and compromise solution. This is in accordance with the value of battery SoC during the start of the outage (See Fig. 11).

The decarbonization metrics ($\mathcal{D}_1$ and $\mathcal{D}_2$) are highest for the decarbonization exclusivity method and lowest for the resilience exclusivity method for both scenarios. This is consistent with the decarbonization app's goal of reducing the usage of dirty energy sources (grid power and fossil-fueled DGs) and maximizing the use of renewable energy. The lowest value of $\mathcal{D}_1$ for the resilience exclusivity method results from the resilience app making setpoint requests to charge the batteries to maximize the amount of reserve energy in the system. However, this is done without considering the source of power used to charge the batteries. This results in fully charged batteries before the peak renewable generation period. Because of this, $\mathcal{D}_2$ is the lowest for resilience exclusivity as surplus renewable power is not fully exploited. Conversely, as the decarbonization exclusivity method implements the setpoint determined by the decarbonization app, it allows for better utilization of renewable energy by charging the batteries using surplus power.

VI. CONCLUSION

This paper introduced an architecture and set of numerical methodologies that can be used to deconflict setpoint commands issued by separate power applications. The structure and elements of the Deconfliction Pipeline were introduced as a method to use standards-based messaging and interfaces to intercept controllable device setpoint requests, identify conflicting setpoints, and determine an acceptable set of compromise values. Three alternative numerical deconfliction methodologies were introduced, based on device-focused heuristics, cooperation between applications, and multi-objective optimization. These methodologies can be conceptually used independently or as components of a broader, globally-optimal application coordination framework. It is anticipated that the architecture and methodologies introduced could be implemented across any ADMS platform with centralized, distributed, or decentralized paradigms to deconflict "black-box" proprietary applications, "naive" power applications that repeatedly issue conflicting setpoint requests, and "intelligent" power applications that share deconfliction-related parameters in a cooperative manner.

Future work will focus on the numerical formulations, data interfaces, and standards-based implementation of each deconfliction methodology in an open-source ADMS platform that enables applications to share coordination information using existing industry standards (such as the Common Information Model). The development of detailed formulations of each methodology will also enable comparisons of their effectiveness for deconflicting applications beyond the initial numerical results presented.

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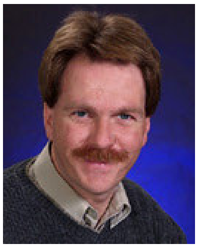

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